

TIDE: Trend-Informed Delayed-label Eviction

Benchmark results

Cache replacement project

September 28, 2026

Summary

TIDE is an ARC cache in which a $12 \rightarrow 64 \rightarrow 1$ MLP, trained online from delayed reuse-distance labels, picks the victim among ARC’s chosen list’s 4 LRU keys, its MRU key and (on a new-key miss) the incoming key. A shadow ARC hands control back when TIDE falls behind. This run compares it with LRU, LFU, ARC, S3-FIFO, SIEVE, LIRS and three learned policies (LeCaR, Cacheus, LRB; libCacheSim reference implementations) on synthetic and real traces.

- **Synthetic traces** (43 trace/size groups, 2 generator instances, 3 TIDE seeds): TIDE ranks #1 with mean normalised hit rate 0.368, against 0.305 (ARC), 0.296 (Cacheus), 0.282 (S3-FIFO) and 0.262 (LIRS). Its mean hit rate is +2.19 points above ARC [+2.12, +2.26], +1.89 above S3-FIFO [+1.82, +1.96] and +0.45 above LIRS [+0.38, +0.52].
- **Real traces** (MSR, Twitter, Meta CDN; first 1,000,000 requests each): TIDE ranks #2. Its mean hit rate is −0.39 points against ARC [−0.43, −0.34] and +0.41 against S3-FIFO [+0.36, +0.45]. TIDE is at least as good as ARC in 1 of 9 settings; worst −1.13 (Meta CDN reag, $c = 38143$). The model was active for only 4–70% of requests because the shadow-ARC fallback kept switching it off.
- **Speed** (per request, 1% cache): p99 falls from 133–219 μ s (NumPy) to 87–112 μ s (Numba, inline training) and 18–26 μ s (Numba, asynchronous training); median 2.9–9.4 μ s against ARC’s 0.16–0.79 μ s. Throughput is 1.2–2.2 $\times$ the NumPy version and still 10–16 $\times$ below ARC.
- **Asynchronous training** costs at most 0.17 points on four of five traces, but −3.10 on phases.

Synthetic traces

Policy	Normalised [95% CI]	Hit %	Wins	TIDE – policy, points [95% CI]
TIDE	0.368 [0.363, 0.372]	44.85	28	—
ARC	0.305 [0.304, 0.306]	42.66	6	+2.19 [+2.12, +2.26]
Cacheus	0.296 [0.293, 0.299]	43.06	9	+1.79 [+1.70, +1.88]
S3-FIFO	0.282 [0.281, 0.283]	42.97	18	+1.89 [+1.82, +1.96]
LIRS	0.262 [0.260, 0.264]	44.40	10	+0.45 [+0.38, +0.52]
SIEVE	0.228 [0.225, 0.230]	42.23	6	+2.62 [+2.54, +2.69]
LeCaR	0.053 [0.052, 0.055]	38.86	4	+6.00 [+5.92, +6.07]
LRB	0.038 [0.037, 0.039]	38.71	3	+6.14 [+6.07, +6.21]
LRU	0.000	38.04	3	+6.81 [+6.74, +6.88]
LFU	−1.235 [−1.268, −1.203]	35.01	6	+9.85 [+9.72, +9.97]

Table 1: Ranking on synthetic traces. Normalised hit rate is $(r - r_{\text{LRU}})/(r_{\text{Belady}} - r_{\text{LRU}})$. Wins count cells with the best hit rate (ties count for every tied policy). Intervals are hierarchical bootstrap CIs over instances and seeds.

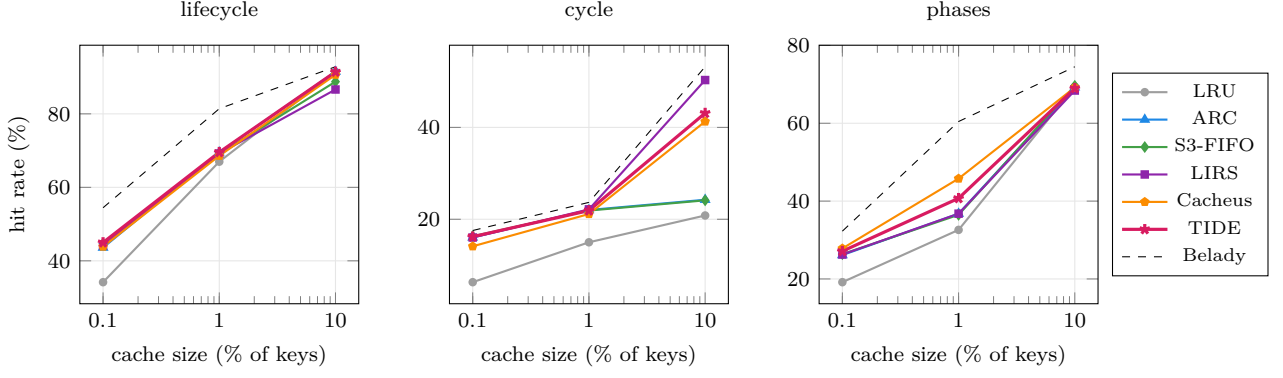


Figure 1: Hit rate against cache size on the trend traces (instance 0).

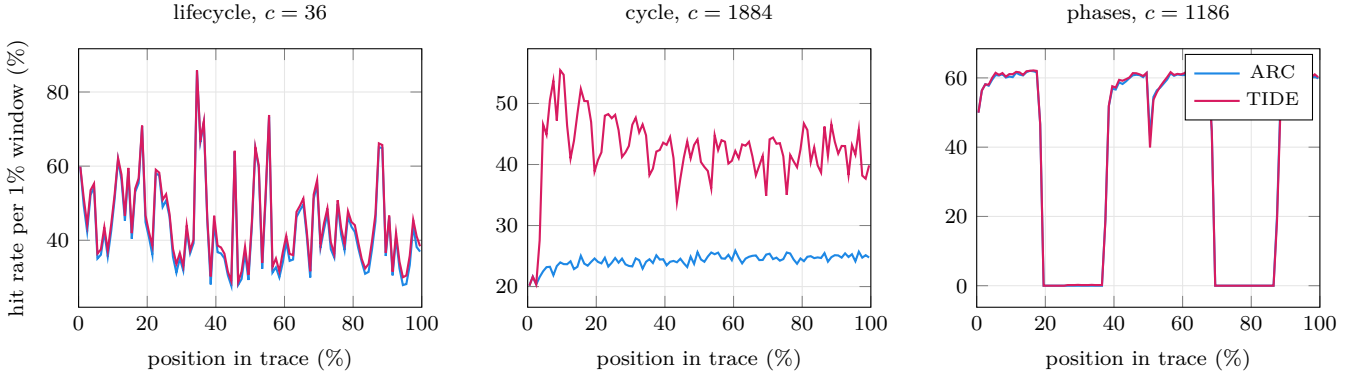


Figure 2: Hit rate over time for ARC and TIDE (seed 0, instance 0) at the size where TIDE gains most on each trend trace.

Real traces

Trace	Domain	Requests	Distinct keys
MSR hm_0	block storage	1,000,000	155,601
Twitter c50	in-memory key-value	1,000,000	109,456
Meta CDN reag	CDN	1,000,000	381,431

Table 2: Real traces (libCacheSim oracleGeneral collection), first requests only; unit object sizes, object hit rate.

Policy	Normalised [95% CI]	Hit %	Wins	TIDE – policy, points [95% CI]
ARC	0.179	53.40	2	−0.39 [−0.43, −0.34]
TIDE	0.113 [0.104, 0.121]	53.01	1	—
LeCaR	0.061	52.55	0	+0.46 [+0.42, +0.50]
S3-FIFO	0.052	52.61	4	+0.41 [+0.36, +0.45]
LRU	0.000	51.72	2	+1.29 [+1.25, +1.34]
Cacheus	−0.005	51.40	0	+1.61 [+1.57, +1.66]
SIEVE	−0.098	51.19	0	+1.82 [+1.78, +1.86]
LRB	−0.163	50.77	0	+2.24 [+2.20, +2.28]
LIRS	−0.388	48.92	0	+4.10 [+4.05, +4.14]
LFU	−1.572	37.51	0	+15.51 [+15.47, +15.55]

Table 3: Ranking on the real traces (one instance each; intervals reflect TIDE seed noise only).

Trace	Size%	c	LRU	ARC	S3-FIFO	LIRS	Cacheus	LRB	TIDE	Oracle	Belady	Δ ARC	Active%
Meta CDN reag	0.1	381	49.40	50.08	49.50	46.48	49.02	49.40	49.95	51.92	55.30	-0.14	16
	1	3,814	53.60	54.42	55.02	52.97	53.23	53.60	54.07	55.57	59.91	-0.35	8
	10	38,143	58.66	58.92	59.15	58.01	58.78	58.66	57.79	59.35	61.86	-1.13	31
MSR hm_0	0.1	156	50.82	54.77	54.74	53.97	51.84	52.77	54.65	57.02	59.64	-0.12	61
	1	1,556	61.23	64.53	63.95	63.96	62.41	63.41	64.54	66.56	69.91	+0.02	53
	10	15,560	71.66	74.08	74.22	73.85	72.52	71.68	73.76	74.25	79.20	-0.32	35
Twitter c50	0.1	109	24.75	24.74	16.13	1.30	24.73	11.11	24.50	26.44	31.72	-0.24	70
	1	1,095	28.57	33.41	35.07	24.40	26.76	30.68	33.03	36.56	51.08	-0.38	4
	10	10,946	66.77	65.62	65.69	65.31	63.30	65.67	64.83	66.42	81.03	-0.79	23

Table 4: Hit rate (%) on the real traces. Active% is the share of requests on which TIDE’s model, not ARC, made the decision.

Speed and asynchronous training

Trace	Implementation	Req/s	p50 (μ s)	p99 (μ s)	p99.9 (μ s)
MSR hm_0	ARC	2,487k	0.16	1.0	3.1
	TIDE, NumPy	128k	2.56	133.5	183.3
	TIDE, Numba sync	150k	2.49	87.3	134.3
	TIDE, Numba async	155k	2.90	25.8	64.1
Twitter c50	ARC	1,099k	0.79	1.8	4.3
	TIDE, NumPy	48k	14.15	219.3	410.8
	TIDE, Numba sync	83k	9.62	112.2	125.2
	TIDE, Numba async	108k	9.43	20.8	58.2
Meta CDN reag	ARC	1,649k	0.28	1.0	5.0
	TIDE, NumPy	81k	7.16	186.8	221.9
	TIDE, Numba sync	122k	4.99	95.9	110.6
	TIDE, Numba async	145k	5.12	17.5	62.7

Table 5: Per-request latency of every access (hits included), measured serially at the 1% cache size.

Trace	Sync hit %	Async hit %	Change	Async steps (% of sync)
cycle	21.90	22.07	+0.17	99
Meta CDN reag	54.07	54.04	-0.02	100
MSR hm_0	64.54	64.44	-0.10	100
phases	40.72	37.63	-3.10	104
Twitter c50	33.03	33.04	+0.01	100

Table 6: Hit-rate cost of asynchronous training at the 1% cache size (mean over seeds).

Caveats

- This is the reduced ($\approx$ 10-minute) configuration: 2 generator instances, 3 TIDE seeds, 500k-request trend traces and 1M-request prefixes of the real traces. LRB in particular needs long histories and may be under-trained on short traces.
- Confidence intervals measure run-to-run noise (instances and seeds) for this set of workloads, not how the results would generalise to other workloads.
- Real traces are evaluated with unit object sizes (object hit rate); byte hit rate is not measured.
- LFU is the standard in-cache LFU; LFU* (global counts, unbounded metadata) is recorded in the CSVs but left out.

Trace	Size%	c	LRU	ARC	S3-FIFO	LIRS	Cacheus	LRB	TIDE	Oracle	Belady	Δ ARC
adversarial	10	16	0.00	0.00	0.00	48.38	0.00	0.00	48.55 $\pm$ 1.94	51.60	49.99	+48.55
bursty	0.1	16	84.32	84.32	84.32	84.32	84.33	84.32	84.25 $\pm$ 0.17	84.76	85.04	-0.08
	1	80	84.43	84.42	84.42	84.42	84.43	84.43	84.26 $\pm$ 0.18	85.22	86.08	-0.16
	10	801	85.52	85.51	85.50	85.51	85.47	85.52	85.19 $\pm$ 0.24	87.04	89.36	-0.32
cycle	0.1	19	6.30	16.03	16.32	16.03	14.41	12.37	16.23 $\pm$ 0.06	16.93	17.57	+0.19
	1	188	15.03	22.05	21.92	22.22	21.26	19.11	21.93 $\pm$ 0.25	22.25	23.72	-0.12
	10	1884	20.87	24.29	24.15	50.36	41.57	37.73	42.71 $\pm$ 0.88	43.71	53.21	+18.43
gaussian	0.1	16	0.26	0.44	0.48	0.45	0.27	0.26	0.41 $\pm$ 0.08	5.12	6.22	-0.03
	1	88	1.50	1.97	2.05	2.01	1.72	1.50	1.94 $\pm$ 0.05	10.86	15.88	-0.03
	10	880	14.78	16.79	17.15	16.99	16.48	14.78	17.97 $\pm$ 0.10	32.55	47.55	+1.18
lifecycle	0.1	36	33.14	42.74	43.90	43.32	43.07	33.14	44.17 $\pm$ 0.93	48.68	53.81	+1.43
	1	359	66.61	69.19	68.81	68.85	68.27	66.61	69.22 $\pm$ 0.44	73.28	81.27	+0.04
	10	3592	91.41	91.44	88.72	86.54	90.59	91.41	91.31 $\pm$ 0.20	91.46	92.75	-0.13
periodic	0.1	16	10.27	11.64	12.39	12.45	11.28	10.27	12.62 $\pm$ 0.10	32.80	36.66	+0.97
	1	89	51.00	66.80	63.79	67.84	67.03	51.00	67.31 $\pm$ 0.39	72.11	76.36	+0.51
	10	891	80.11	80.03	79.60	78.54	79.88	80.11	79.73 $\pm$ 0.17	81.94	85.56	-0.30
phases	0.1	119	19.17	26.17	26.41	26.30	27.88	20.23	27.16 $\pm$ 0.65	29.20	32.26	+1.00
	1	1186	32.57	36.77	36.50	36.79	46.58	33.21	40.87 $\pm$ 3.60	53.12	60.44	+4.09
	10	11861	69.42	69.26	69.61	68.45	69.13	69.42	69.01 $\pm$ 0.07	70.21	74.50	-0.25
uniform	0.1	16	0.15	0.17	0.17	0.17	0.14	0.15	0.16 $\pm$ 0.02	3.75	4.82	-0.01
	1	100	1.01	1.00	1.01	1.03	0.98	1.01	1.02 $\pm$ 0.04	8.56	13.17	+0.02
	10	1000	10.04	10.14	10.05	10.07	9.96	10.04	10.04 $\pm$ 0.10	25.42	40.05	-0.09
web	0.1	16	3.39	4.52	5.18	4.88	3.93	3.39	4.86 $\pm$ 0.09	18.73	21.59	+0.34
	1	50	10.49	14.18	14.87	14.89	12.80	10.49	14.96 $\pm$ 0.04	31.43	38.99	+0.77
	10	499	67.74	71.35	70.79	70.73	70.44	67.74	71.04 $\pm$ 0.07	73.04	78.85	-0.32
zipf-100k	0.1	19	35.46	47.67	49.27	48.18	47.59	35.46	49.25 $\pm$ 0.05	51.22	53.53	+1.58
	1	194	59.43	67.28	67.35	67.45	67.48	59.43	67.57 $\pm$ 0.03	68.19	71.26	+0.29
	10	1935	74.08	77.00	76.65	77.15	76.85	74.08	77.24 $\pm$ 0.13	78.22	80.16	+0.24
zipf-a0.5	0.1	16	0.38	2.20	2.66	2.22	0.68	0.38	2.36 $\pm$ 0.09	6.26	7.20	+0.16
	1	100	2.39	6.58	7.35	6.70	3.15	2.39	7.20 $\pm$ 0.09	13.04	18.26	+0.61
	10	999	17.98	23.67	25.05	23.80	22.76	17.98	25.77 $\pm$ 0.10	32.38	48.12	+2.10
zipf-a0.8	0.1	16	4.29	13.63	14.87	13.79	12.02	4.29	14.53 $\pm$ 0.22	17.92	19.90	+0.91
	1	97	15.50	26.21	27.14	26.48	25.94	15.50	27.35 $\pm$ 0.20	30.18	36.63	+1.14
	10	970	43.06	50.34	51.46	50.41	51.19	43.06	51.82 $\pm$ 0.12	54.17	65.75	+1.48
zipf-a1.0	0.1	16	17.98	30.75	32.75	31.20	30.15	17.98	32.75 $\pm$ 0.15	35.77	38.35	+2.00
	1	85	37.12	48.12	48.97	48.45	48.65	37.12	49.27 $\pm$ 0.11	51.04	56.49	+1.15
	10	853	65.33	70.72	71.27	70.77	71.47	65.33	71.43 $\pm$ 0.09	72.20	79.40	+0.71
zipf-a1.2	0.1	16	33.30	45.70	47.51	46.18	45.69	33.30	47.57 $\pm$ 0.09	49.72	51.83	+1.88
	1	89	52.82	61.74	62.30	61.96	62.19	52.82	62.42 $\pm$ 0.11	63.31	67.02	+0.68
	10	886	72.01	75.57	75.84	75.64	75.77	72.01	75.83 $\pm$ 0.10	76.88	82.21	+0.26
zipf-a1.5	0.1	16	72.58	77.96	79.82	78.63	79.05	72.58	80.02 $\pm$ 0.14	81.21	82.89	+2.06
	1	28	79.19	83.40	84.46	83.86	84.29	79.19	84.61 $\pm$ 0.08	85.48	87.28	+1.21
	10	282	93.37	94.68	94.69	94.71	94.74	93.37	94.75 $\pm$ 0.02	94.99	95.95	+0.07

Table 7: Synthetic results per trace and size (mean over instances; TIDE with a 95% t-interval over instances $\times$ seeds). Bold marks the best contender. Oracle is TIDE with true reuse distances; Δ ARC is TIDE - ARC.